# HDOC Day-68

**Question:** Write a recursive function to convert a decimal number into its binary representation.

## Python Program

```
def decimal_to_binary(n):
    if n > 1:
        decimal_to_binary(n // 2)
    print(n % 2, end="")

num = int(input("Enter a decimal number: "))

if num < 0:
    print("Please enter a non-negative number.")
else:
    print("Binary representation: ", end="")
    decimal_to_binary(num)
    print()
```

## Sample Output

```
Enter a decimal number: 10
Binary representation: 1010
```

## Explanation

The function calls itself with the number divided by 2 until the number becomes 0 or 1. As the recursive calls return, each remainder (0 or 1) is printed in the correct order. For example, decimal 10 is represented as 1010 in binary.

## Conclusion

This program uses recursion and the remainder operator to convert a non-negative decimal integer into its binary representation.